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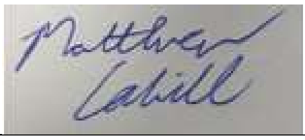

School of Engineering

“A review of the literature in the field of onshore wind turbine high fidelity aeroelastic modelling techniques”

2024

Compiled by Matthew Cahill

as part of master’s in mechanical engineering

Abstract

Aeroelastic modelling has become a significant tool for wind turbine analysis, as sizes grow, and further optimisation is required, more detailed simulations that consider complex effects must be developed. This paper presents the tools and techniques used in high-fidelity aeroelastic modelling, including CFD in the aeroelastic components, and FEM methods in the structural component as well as their coupling. This paper concludes on a gap in the research on the prohibitive costs of high-fidelity methods, and a proposed 36-month research study to address these issues.

Contents

1. Introduction

Wind energy is vital to achieve the 2050 net-zero goals of the Paris climate agreement, wind energy is abundant, and with offshore wind, highly available. However, offshore wind is currently challenging to implement, and floating offshore wind is still in its infancy, however it does hold the potential for even larger 20 MW turbines. For them mean time; while floating offshore technology matures, onshore wind is a viable stopgap.

However, the challenge with onshore wind, is space. Building onshore wind turbines is often contested, by local communities, or environmental challenges, or by difficult terrain. Thus its advantageous to optimise the design of turbines as much as possible, to maximally utilize the space that can be used.

However, the challenge then arises, to design bigger and more efficient turbines, we need more advanced engineering simulations, as the current industrial methods, such as BEM and actuator models, are no longer cutting it. Aeroelastic modelling is the tool used for wind turbine design, as it enables the analysis of the complex interactions between aerodynamic forces and structural movement and deformations. Aeroelastic modelling works by computing the aerodynamic forces along the wind turbine, which gets passed to a structural deflection model to compute the elastic deformation of the blades, which is then passed to an FEM model to analyse stresses, the structural deformations are then returned to the aerodynamic model, to recompute the aerodynamic effects as the deflection alters the aerodynamics, and the cycle repeats again, thus considering effects of turbulence, and dynamic wind behaviours, it is a very complex interaction to model.

Numerous attempts have been made over the years to preform aeroelastic design. It began as a tool for aerospace design. Aeroelasticity can trace its roots to Caltech, under Von Karman and Edwin Sechler, and since numerical and analytical tools have been developed for its application. In terms of wind turbines, the current field uses techniques such as BEM and Actuator models to mathematically analyse the aerodynamic effects of flow. However, these models require continuous adaption, improvements and add-ons and engineering corrections to account for increasingly complex flow behaviours. These so called 'low fidelity' models do not account for behaviours such as dynamic inflow, and are designed under quasi-static assumptions, however the wind does not care for our neat engineering simplifications. New turbine designs need to account complex flow behaviours such as, dynamic stall, tip vortices, turbulence induced vibrations, dynamic rotor position controls, tower effects, advanced materials, wake fields, etc. Aswell, as turbines grow to 20 MW size, blades grow larger, they become more flexible, the effects of blades bending has become significant, even reaching the point where they can flex to hit the tower, this has the effect of inducing significant aerodynamic effects, significant stresses on the blades and induces vibrations in the blades that could lead to catastrophic effects. These lot of these challenges require significant overhauls and additions to existing mathematical models and may not even be possible to capture.

In their stead, CFD and FEM methods have shown to pick up where the lower fidelity models fall short. By their nature, they can capture the complex behaviours that previous models struggle to do. Previously, the computational demand of these so called 'high fidelity' methods were prohibitive, and so BEM methods were preferred by industry, but with the growth in both computing power, and optimisations to CFD and FEM methods, the high-fidelity methods are becoming viable for cases of wind turbine design and optimisation.

The field of high-fidelity modelling is still maturing, and there exists significant scope for improvement. This literature review details the components of high-fidelity modelling and the current state of the field, it is concerned mainly with onshore wind turbine design, however sources draw from floating offshore wind turbine design, where high fidelity methods are also of great interest due to effects of the sea, as well as aerospace and maritime engineering papers.

Aeroelastic modelling consist of three main components, the aerodynamic model, the structural model and their coupling methods, detailed in Figure 1. The aerodynamic models use lower fidelity methods such as BEM and actuator methods, however this review concerns the higher fidelity CFD methods. The structural model can use the high-fidelity direct FEM method, or the 1D beam method combined with FEM methods, which would till fall under high fidelity, which id detailed in this review.

This review proceeds plans for a 36-month engineering study on high fidelity aeroelastic modelling, and its objective, is to understand the current state of the art in the field, and identify a research gap, that the study can address.

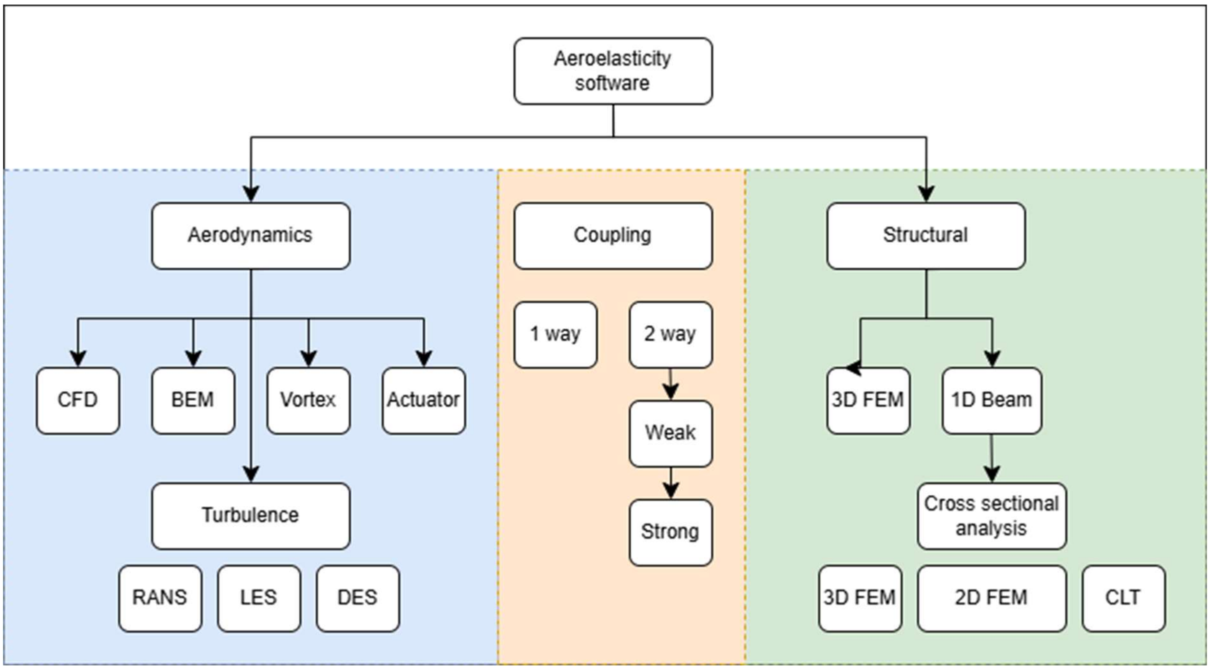

Figure 1: Aeroelastic modelling flow chart

2. Methodology

To ensure quality of information, care was taken when choosing the papers selected for this review. The journal impact factor of their parent journals, and the papers citations were considered before use. In some cases, papers were taken from respected bodies, such as NASA or ASME. Google scholar was used for the paper search, as well as tools such as Litmaps[1] to find key papers that others draw upon and cite. In terms of searching, google scholar and Scopus were used, and keywords such as CFD, FEM and other always combined with “wind turbine aeroelastic modelling” were used to ensure relevant papers were found.

3. Aerodynamic Model

The first component of aeroelastic modelling is the aerodynamic model, which concerns the loads exerted on the blades from airflow, the turbulence effect after crossing the blades, the turbulence vortices at the blade tips and the wake field left behind. When the aerodynamic loads are computed for timestep i, the values can be passed to the structural model for that timestep to calculate deflection. The new deformed structure is taken and then passed back into the aerodynamic model for timestep i+1 to compute the loads for timestep i+1.

Four classes of aerodynamic models exist with varying levels of fidelity, BEM (blade element momentum), Vortex models, Actuator type models and computational fluid dynamics (CFD both FVM and FDM), there exists hybrid models that contain parts from each. As the effects of turbulence gain significance in design, such as dynamic stall, wake fields, tip effects, turbulence models are added to these aerodynamic models, such as RANS, LENS and DES.

This review however is only concerned with the high fidelity CFD method, which is detailed in the following sections.

3.1.CFD

While BEM is the preferred tool in industrial applications, due to its speed, the growing size of wind turbines, demand for increased optimisation and higher yields is driving the adoption of higher fidelity methods of aeroelastic simulation. Thus, computational fluid dynamics (CFD) methods applied at large scale.

Computational fluid dynamics solves the Navier-stokes equation set and its derivatives using numerical or iterative convergence approaches e.g. Jacobian iteration. The nature of CFD provides more detailed predictions of wakes, time-dependant changes, tower disturbances, dynamic stalls and other phenomenon that BEM and other lower fidelity methods require supplementary models to rectify for.

CFD methods have been shown to provide higher accuracy on simulation results against BEM methods, however, their computational cost and time has proven a to be a limitation on its adoption. The following is a review of CFD methods, their advantages and disadvantages.

3.1.1. Governing equations

Computational Fluid Dynamics (CFD) simulations are based on the Navier-Stokes equations, a set of partial differential equations that describe the conservation of mass, momentum, and energy in a fluid [2]. Compared to the simpler BEM methods, these equations provide high accuracy, able to capture complex flow patterns, and capture the vortex and turbulence effects by nature that BEM requires supplements to do.

These equations are nonlinear partial differential equations, and while it is theoretically possible to solve analytically in some cases, it is not possible in all cases, or it requires inordinate amounts of compute. Analytical discretisation methods have reigned in the field of CFD. These methods discretize the equations by converting them to a set of algebraic equations that can be solved via numerical iteration methods, such as Jacobian iteration[3]. Finite volume, finite element and finite difference schemes can be used, however finite volume methods FVM reign in terms of computation speed and versatility.

These discretization methods are solved iteratively, with the aim to reach a converging solution, by analysing the gradient descent of the flow parameters each step until a target gradient change is reached. The accuracy of the CFD solution depends on factors like the quality of the mesh, the choice of numerical schemes, and the appropriateness of the turbulence model. Validation of the simulation results against experimental data is crucial to ensure the reliability of the CFD model.

3.2.FVM discretization & Meshing

In FVM, the computational domain under study (E.g. Wind turbine and flow tunnel) is divided into a finite number of small control volumes (CVs) (represented by the mesh cells). The PDE form conservation equations are integrated across each CV using the surface and volume integrals, producing a set of algebraic equations. Care is taken to ensuring that the change of the conserved quantities (mass, momentum, energy) across the CV boundaries are balanced by any energy sources (exothermic reaction etc.) or sinks (viscous friction etc.) within the CV. Values on the faces of CVs are obtained by interpolation of the values at the centres of the CVs. Interpolation schemes are a control choice in the simulation setup and can affect the results drastically, such as upwind difference, central difference and their higher order variations, which affect how many neighbouring cells to consider. A FVM mesh is shown in Figure 2, this is taken from [4], to compare, a meshed wind turbine blade is shown in, taken from [5].

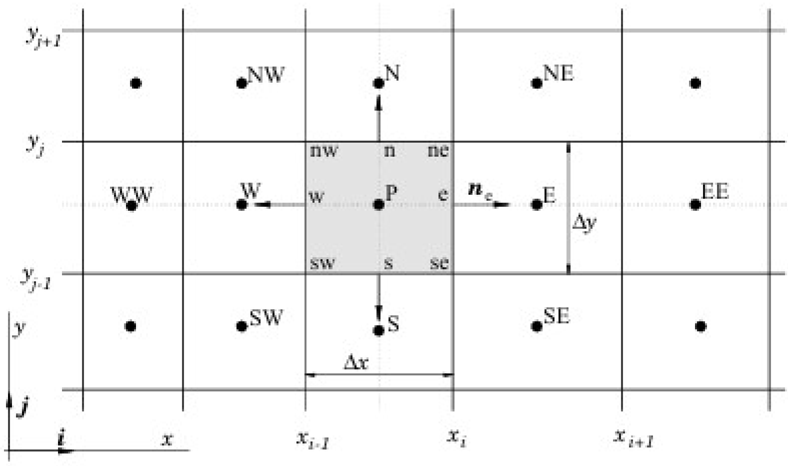

Figure 2: Finite volume method mesh diagram

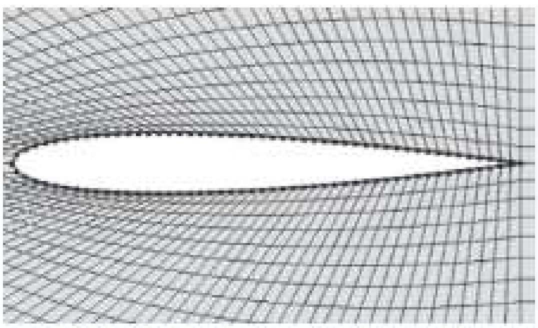

Figure 3: Wind turbine blade with mesh

3.2.1. Meshing

The mesh is one of the most critical aspects of a CFD solver, as it impacts both computational cost and accuracy[6], as the denser the mesh, the larger the equation set to solve, the longer the computation time. As stated above, the mesh elements are the control volumes the governing equations are discretized over. However, there is a limit to how coarse or fine mesh can be, lower resolution coarser meshes may not provide accurate results, however the complexity of calculations grows exponentially with the fidelity of the mesh. Ideally, the coarsest mesh that provides accurate results is used, as it would have the lowest computational cost, therefore sensitivity studies[7][8] [9]must be carried out preliminarily to ensure convergence to experimental results. Figure 4 shows the relationship between mesh size and computation time, taken from [9]was a study of aeroelastic simulations on aircrafts, however the same principles and design systems hold.

3.2.2. Mesh Motion Techniques

A difficulty arises in the case of aeroelastic simulations, in that the mesh for the wind turbine blades is moving relative to the mesh of the surrounding air stream, a selection of techniques exist to solve this:

Sliding Mesh:

Sliding meshes are a technique where two non-overlapping mesh regions slide over each other[7], an interface region is created between the moving and static (blade and free stream), in a timestep the moving mesh rotates with a defined velocity across the static mesh, when the timestep is complete the moving and static vertices are reattached [10]. This technique is generally good in steady state flow and rotation cases [11].

Deforming Mesh:

Deforming meshes are used in CFD to simulate fluid flow around objects that move or change shape, such as aeroelastic deflection of the blades. the mesh deforms with the object eliminating the need to create a new mesh every time step. Various methods are used to perform this deflection, such as the Elastic Analogy (EA) [8] method. EA models the mesh as an elastic solid with variable young's modulus reciprocal to the mesh volume, displacements are calculated accordingly, where small elements act as rigid bodies and large elements absorb deformations, as applied deformations are relatively small [9]

Chimera Mesh:

Chimera meshes, assemble two or more grids into a computational domain by the interpolation of the boundaries. The airflow is considered the background mesh, and the blade the overset mesh. Background mesh cells that overlap with overset cells are removed called "hole cutting" for that timestep. A domain decomposition coupling algorithm is applied to obtain a continuous solution to the equation set across the interface of the two meshes [12]. Chimera meshes are advantageous in the case of wind turbines, where the nature of the design is rotating blades within the mesh.

3.3. Turbulence models

The Navier Stokes equation set in its base form is adept at modelling simple, laminar flow fields, or fields with slight turbulence, however geometries or flow parameters that would induce high turbulence struggle to converge and demand enormous computing power, for this, specialised turbulence models have been developed, to efficiently capture the smaller details of turbulent flows and turbulence induced vibrations[4].

Such examples are large eddy simulations (LES) [4][11] [13][14]which solves large scale eddies directly and models the smaller turbulent eddies using sub grid-scale models [15] LES deals well with large scale turbulence eddies, however smaller eddies and boundary region/wall turbulence suffers. An alternative to LES is the family of Reynolds Averaged Navier Stokes flow models (RANS), which are divided into one equation, and two equation sets, ne example being the K – ε[16], which solves for the turbulent kinetic energy equation *k* and the turbulent energy dissipation rate ε. RANS models deal well with flow near walls. There also exists a family of turbulence models called Detached Eddy Simulations (DES) [11] [15] [4]which are a hybrid of LES and RANS models, leveraging the strengths of both, LES models are used for regions away from walls, and RANS used near walls. It has been shown that DES models perform well in wind turbine applications, capturing flow separating and wake dynamics well. There exists a family of DES modification models, such as Delayed DES (DDES) and Improved Delayed DES (IDDES)[14].

Wind turbine aeroelastic simulations require turbulence effects to be modelled, as tip vortices, wakes, dynamic stall and boundary shedding are all aerodynamic effects, and all impart design necessities.

Type	Family	Model
RANS	One equation	Spalar-Allmaras[17]
RANS	Two equations	K - ε
RANS	Two equations	K - ω
RANS	Two equations	K – ω SST (k- ε/k- ω hybrid)
LES	eddies	[13]
DES	DES	RANS LES Hybrid
DES	DDES	RANS LES Hybrid
DES	IDDES	RANS LES Hybrid

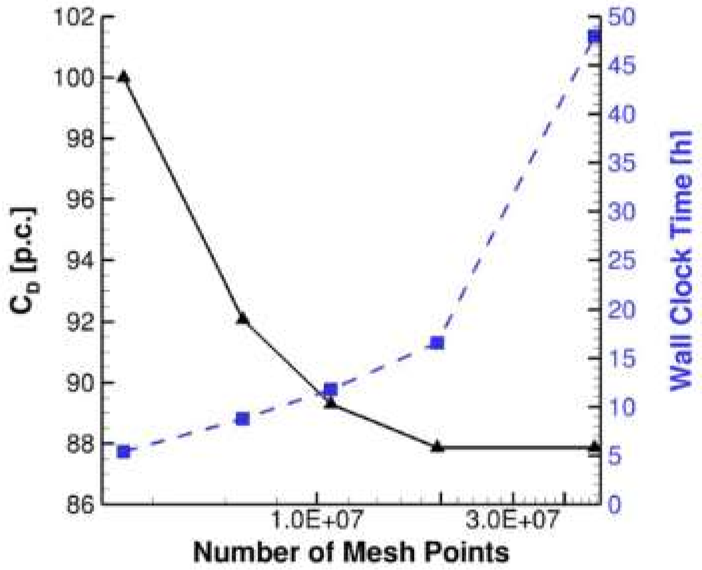

Figure 4: Merle et al: CFD mesh versus calculation time

4. Structural Model

4.1. 3D FEM

3D Finite element models provide the highest accuracy and precision in the results of WT aeroelastic response models, as FEM models with sufficiently detailed meshes can capture stress localities and nonlinear behaviours accurately. The disadvantage to this, is that the computational cost of 3D FEM structural models is very high. For this reason, 3D FEM is considered a high-fidelity model.

In 3D FEM models, the wind turbine blade structure is digitized using composite shell elements[18], which allows for the representation of composite layer characteristics.[5]. A CAD model of the blade is created and software's such as NuMAD 2.0 are used to preprocess the blade before use in a finite element software, which is listed in the software section. Shell elements are used as the composite materials directional properties can be defined as a vector [19], solid element models can be used as well, which may provide more detailed tress behaviours, however they induce complexities in the simulation and require more computing time. Figure 5 shows the cross section of two blade FEM meshes, from [19]. Figure 6 shows the stress contour plot as the result of an FEA simulation from [20]

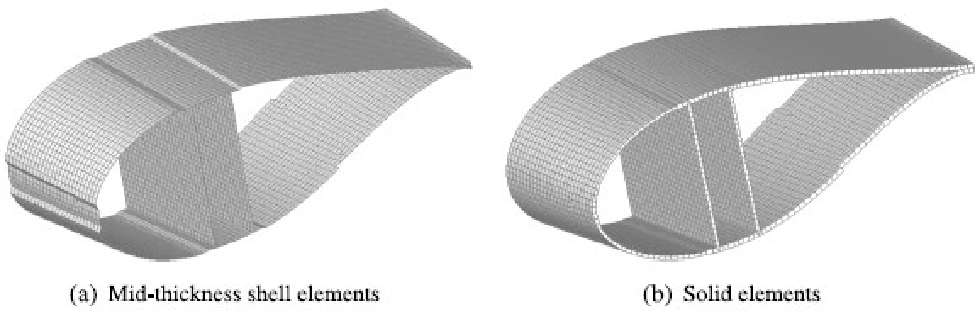

Figure 5: Two FEM models of blade cross sections [19]

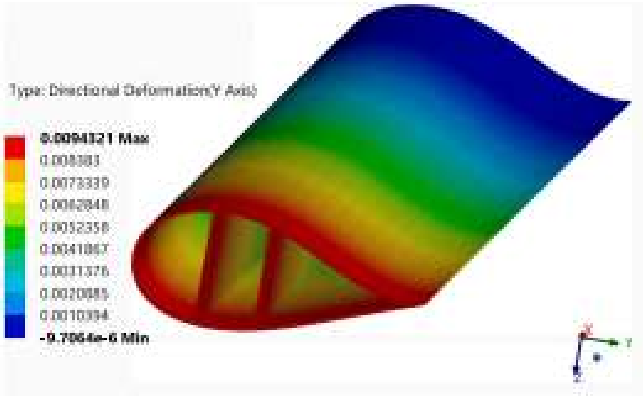

Figure 6: An FEA blade model with stress distribution plotted

4.1.1D Beam

4.1.1. Euler-Bernoulli vs Timoshenko

Due to the cost of 3D FEM models, 1D beam models are often used in its stead. While a lower fidelity class of model, combined with cross section analysis techniques, the 1D beam model can be upgraded to a pseudo higher fidelity. [20]found that there is an error of only 0 to 5% on Euler Bernoulli applied to wind turbine blades, as opposed to FEM methods.

Wind turbine blades are significantly longer than they are thin, thus beam theory can be used to model their structural response behaviour, as the elastic response behaviour of the blade can be lumped into an equivalent simple beam [21][22] Two beam theories are commonplace in WT aeroelastic models, Timoshenko beams and Euler-Bernoulli beams.

Timoshenko beam theory is suited to shorter beams where shear stress forces will dominate, while Euler-Bernoulli beams suit to longer beams with flexure dominated behaviours. Both models agree within mid length beams. So, it stands that the simulation setup should consider this when choosing the model [23]An example of an Euler-Bernoulli beam undergoing torsion is shown in Figure 7, from [20]

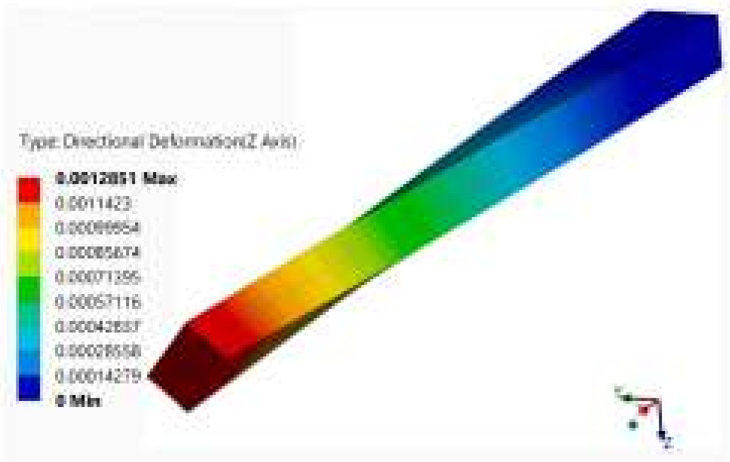

Figure 7: Torsional loading of an Euler Bernoulli beam

4.1.2. Modal

Modal analysis is the process where vibration dynamics and deflections of a WT blade, can be modelled as a linear superposition of the set of natural vibration modes of the blade[5] [24] [14] . Usually, the dominant lower harmonic modes are used, such as the 1st and 2nd flap wise and edgewise models. [25]. Examples of Aeroelastic codes that use a modal approach include FAST[26], FLEX 5 and OpenFAST [27]

4.1.3. MBD

Multi Body Dynamics is an approach that divides the blade into multiple rigid bodies and connects them with joints and springs[28]. In this approach the bodies can be either rigid (strainless) or flexible (generate strains), and the joints which act as kinematic constraints between bodies restricting relative motion. From this, a set of techniques exist to solve the constrained dynamics of the system such as embedding techniques, augmented formulation and the discrete Euler Lagrange equation[29]. MBD is more computationally expensive but a review by found the results to be more accurate than modal approaches. [14] MBD is implemented in HAWC2 [30]and OpenFAST.

4.1.4. 1D FEM

1D FEM discretizes the blade into a series of beam elements connected at by internal forces and displacements at nodes. The beam deformation is computed using the fem method. 1D methods can accurately describe beam deflections[14][28].

5. Cross section analysis

When 1D beams theory is employed, it must be supplemented with composite modelling to capture the material characteristics of the composite layers. Figure 8 shows a cross section of a turbine blade, and an example of a possible composite layer structure making up its skin, from [31]. Wind turbine material design is an integral component, and composite material behaviours are complex and anisotropic. Thus, a selection of methods has been employed in this case, 3D FEM, 2D FEM and Classical lamination theory CLT.

5.1. 3D FEM

3D FEM cross sectional analysis uses a detailed CAD model of the blade and FEM techniques to model the cross-sectional stress. As opposed to direct 3D FEM methods, this method takes the spanwise discretized sections of the 1D beam model and creates equivalent spanwise discretized 3D sections. First the divided 3D FEM sections are meshed, and the anisotropic composite material properties applied. The displacements calculated for each section on the 1D beam model are then applied to the individual respective 3D FEM sections. [32]

This method provides high accuracy as 3D FEM methods can capture detailed stress localities. The advantage to this method, is smaller chunks of the blade can be analysed with FEM, reducing the computation needed if only certain sections are of interest. These are used in the preliminary design phase according to [33], as well as in buckling and fatigue analysis.

5.2. 2D FEM

This method involves using 2-dimensional cross section planes, with the blade divided into spanwise sections, with material design variables, such as skin thickness, shear webs and spar caps defined at these sections, refer to Figure 8. Each section is then modelled with an equivalent 2D cross sectional panel, for which a stiffness matrix is computed. These stiffness matrices are returned to the 1D beam model to inform how it behaves under loading [32]. Some software for this analysis includes: VABS & PreVABS[34] , BECAS & Airfoil2BECAS[35]. Figure 9 from [32] demonstrates a process of 2D FEM blade design optimisation, note on the top right the spanwise sections of the blade.

5.3. CLT

CLT or Composite laminate theory attempts to model the mechanical properties of the turbine blades mathematically. CLT is built upon classical plate theory, which is used to model materials such as carbon fibre, fibreglass or hemp, if you're Porche [36]. This allows for the creation of a directional stress matrix, as these materials are anisotropic, and they are strongest in the direction of the fibres, and often have a layup so the output is a function of their individual components.[37] [28]. This methods advantage is that it is more efficient to compute, at the drawback that modelling the shapes can be difficult and inaccurate, and complex stress localities can be lost.

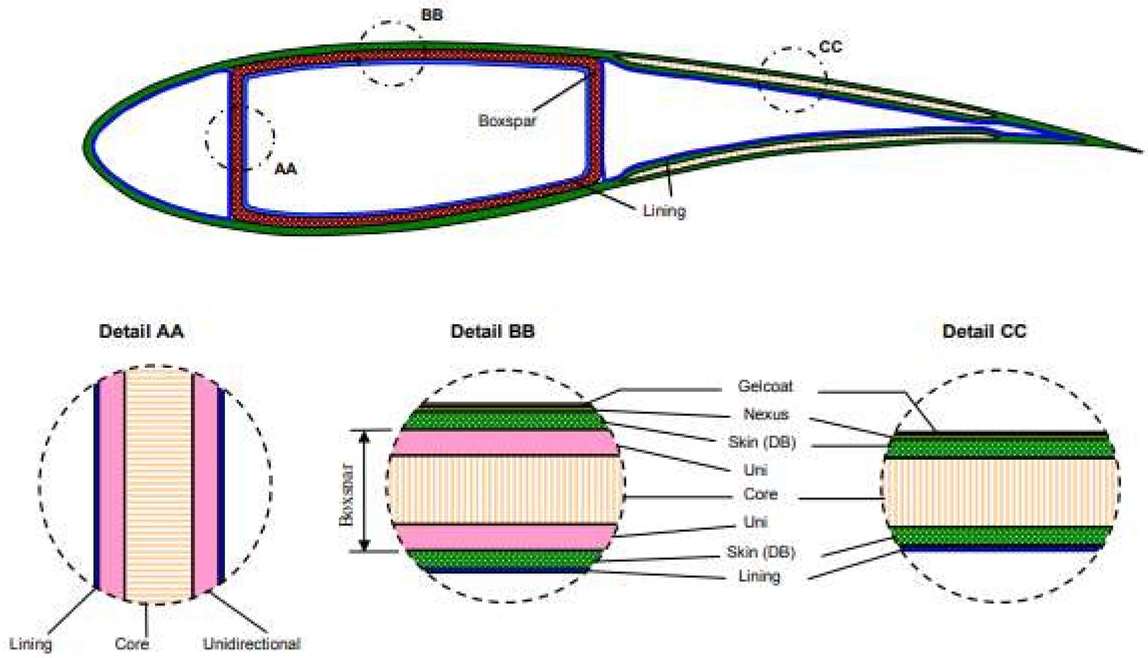

Figure 8: Anatomy of a wind turbine blade and its composite layers

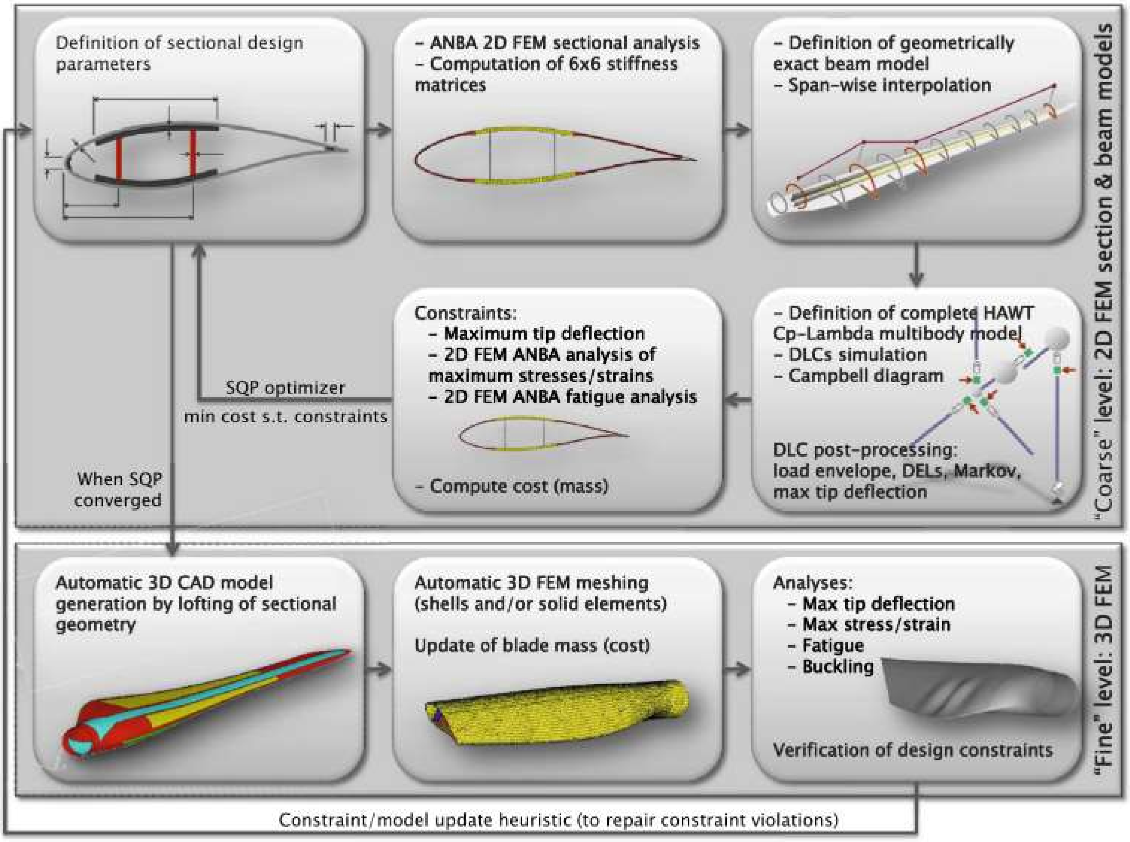

Figure 9: Flowchart describing 2D FEM analysis

6. Coupling Methods

Fluid structure interface or FSI coupling is the method by which the aerodynamic forces on the turbine are connected to the structural deformations and vice versa. Two methods exist: One-way coupling and two-way coupling. One way coupling ignores the deflection of the rotor blade in the aerodynamic model, only passing aerodynamic data to the structural model, but not vice versa. This review is concerned with high-fidelity models, which this does not fall under. [38] Reports on a case study where the two methods were compared, concluding that two-way coupling is superior, accounting for tip deflections, one way coupling induced damping.

6.1. Two-way coupling

Two-way coupling is the method by which the aero forces are passed to the structural forces and the structural deflections are passed to the aero model. Two methods of performing this exist, weak and strong coupling. Weak coupling is where the fluid and structural solvers are independent components, and data is exchanged at a defined interval, such as every time step. Strong coupling is where the two modules exchange data more frequently, withing a single time step or are solved simultaneously. [4][39][38]. A methodology for simulating the full rotor, nacelle and tower with FSI was developed in 2014 [40], shown in Figure 10, which found that the interaction between the tower has significant aerodynamic effects, and a drop in torque is discovered as the blade passes the tower, therefore future high-fidelity models are recommended to use this.

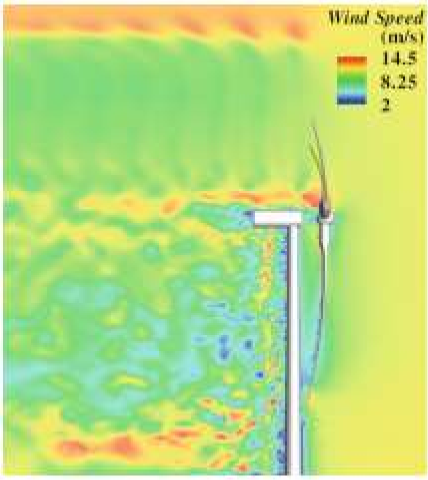

Figure 10: Aerodynamic effect of the tower

7. Findings & Conclusions

Following a review of the literature, the key findings are as follows. The drawbacks of previous mathematical methods, such as BEM and actuator models, is that they require supplementary models to capture effects such as tip vortices, and dynamic stall, and inherently assume simple, quasi static conditions, CFD however by nature, captures these, and the accuracy of the simulation is in the input conditions, and not on the limitations of the models. FEM structural methods are the most accurate, be it direct 3D fem, or 1D beam FEM combined with 2D or 3D cross sectional analysis techniques. These provide detailed stress plots, that capture complex and minute stress behaviours, and accurately model the deflections and vibrations induced. Strongly connected two-way coupling methods provide the most accurate representations of behaviours, and enable the capture of nonlinear effects, which is required as blades get larger and so too deflections.

The drawback found however, is that these high-fidelity methods are computationally expensive, to the point where it is prohibitive, as lower fidelity methods are still used in commercial design optimisation due to their budgets and deadlines, two weeks for one simulation is unrealistic [28]. As stated by Sebastian, Perez and Becker 2021 [41], "*Even if we introduce techniques that sacrifice details of the flow to combine different scales, the computational power required to do aeroelastic load calculations of wind turbines using full flow field CFD-methods is too high for these methods to be solely used in the design and certification*", and supported by However, a gap these papers either did not address, or cited for future work, is the use of machine learning and deep learning methods to reduce computational complexity and time.

As a side note, this paper is not a wind turbine investigation, but an aeroelastic, aero control surface investigation for a plane design study, [42] Reinbold et al 2022 required 2250 CPU hours for an aeroelastic simulation, equal to 13.4 weeks of computation time for one simulation. To put simulation times into perspectives.

8. Research Study

8.1. Research Gap

The need for wind power grows, so too does the need for larger turbines, and for faster design cycles and better control algorithms. However, challenges that limit the speed at which simulations can be performed arise, for example dynamic inflows, larger blades having significant deflections, wake analysis, tower effects, strong coupling methods etc.. increase the computational cost. Currently the computational cost of high-fidelity methods (CFD FEM) is prohibitive, and lower fidelity methods are preferred in industrial uses. Case studies exist in the use of machine and deep learning methods to reduce the computational complexity of CFD and FEM methods, and there is a handful of tools available. The suitability of these tools and models, and the development of novel applications for wind turbine aeroelastic modelling could prove valuable, as it would reduce the current barriers to entry of high-fidelity methods. However, limited studies exist, and large amounts of clean and suitable data is needed.

8.2. Research Purpose

The purpose of this research study is to use existing machine/deep learning tools developed for CFD and FEM applications that have proven to be successful, and to investigate their integration into wind turbine aeroelastic modelling pipelines. The goal is to reach a significant reduction in computational time, allowing for the use of more complex models.

8.3.Research questions

1. Can Machine learning – CFD solver coupling tools, used to reduce convergence time be applied to aeroelastic simulations
2. Can deep learning models provide a more accurate turbulence model, that compensates for the downfalls of LES and RANS models and allows for more accurate simulation of blade turbulence effects such as tip vortices, dynamic stall and boundary layer separations.
3. Can CFD - machine learning super-resolution augmentation methods be used to speed up simulation time by allowing the use of a coarser mesh
4. Can computationally cheaper surrogate FEM models such as smart elements be used to reduce the computation time of the structural deflection components, without sacrificing accuracy.

8.4.Research objectives

1. Machine learning CFD coupling tools have been shown to reduce convergence time, such as CFDnet [43], which demonstrated speed ups of convergence time of 2.3 to 3.1 times on aerofoils in turbulence. The first research objective is to take an existing wind turbine aeroelastic design study with good data, including compute time, and replicate the study using CFDnet to speed up convergence time. The study should then investigate the speed up achieved, and the accuracy of the results for the base CFD versus CFDnet.

2. Turbulent flow is one of the more difficult components to simulate, and several models have been developed as mentioned in section 3.3, such as RAND and LES. Case studies exist of turbulence neural network models being developed for specifically LES wall modelling and its applications [44][45], which proves viability for this aspect. The objective of this component is to build, test and validate a novel turbulence model from existing turbulence data which can be implemented into CFD software's and serve as a tool to more accurately capture turbulence effects at different eddy scales for future design studies.
3. Super resolution machine learning methods can take a low-resolution data field, such as an image, or a turbulent flow map, or an FEM stress contour grid, and augment the data to a higher resolution scale, by means of supervised training a model (such as a multi-layer perceptron model or a convolutional autoencoder) on the data upscale process. A case study is presented on the use of this method to upscale vortical flow fields [46]. The objective of this step is to collect a dataset of wind turbine turbulence fields produced by simulations, at a low (testing data) and higher resolution (for a ground truth), build a super encoder model, and investigate the accuracy of the super resolution encoder at upgrading the lower resolution data to a higher resolution and comparing the ground truth higher resolution set.
4. Machine learning methods have been demonstrated to be able to replicate the results of finite element simulations, without the need to compute internal node displacements, such as the smart element approach, presented in [47], which demonstrated the ability to model structural deflection from imposed forces, without the need of iterative or internal displacement methods, thereby reducing the computational cost and increasing computation speed. This component of the study should focus on implementing smart elements in blade FEM simulations. Due to the smaller scale this is capable of, it is likely not ready for full connected aeroelastic sims, but a trial on an isolated blade model with deflections applied, and comparing to traditional FEM would provide a good case study.

8.5.Risk assessment & Contingency plan

Risk 1: Data availability

The studies cited in the research objective mention the need for clean noise free data, which is vital for training machine learning models, also models tend to require large amounts of data, proportional to the model's depth and complexity. While a sizeable library of wind turbine aeroelastic design studies exists, access to the raw simulation output results required for training may be limited or unavailable.

Contingency plan 1:

By developing or obtaining an existing high fidelity aeroelastic model, the required result data can be produced in house and used for the study, this allows for better control of data quality, and for obtaining the different classes of data required, such as flow maps, and FEM deflections.

Risk 2: Disease/virus outbreak requiring quarantine periods

The covid 19 epidemic demonstrated that research and daily activities can be derailed due to the need for quarantines and lockdowns, preventing research teams from accessing labs together etc. In the case another similar epidemic arose, without plans in place the entire study could get halted.

Contingency plan 2:

The contingency plan in this case is to have a robust workflow setup, with cloud access to collaborations software (e.g. Teams) data and supercomputer/GPU clusters, so that if quarantine is required, researchers could work from home, and remotely access the required data and powerful computers in safety and collaborate over the internet.

Risk 3: Scope too large

Development of new tools could quickly lead to unnecessary and large amounts of work, developing supporting tools and pipelines, which would put the project off track.

Contingency plan 3:

The research goals chosen are based upon existing and successful tools, that have been built to fit into existing software's and pipelines, such as ANSYS etc. Care should be taken in the initial development/acquisition of tools to keep them within scope and integrate into existing software to reduce development complexity/times/costs.

9. Research Gantt chart

10. Referencing tools

Mendeley cite, the citation tool built directly into word was used for citing references within the text, all papers cited were added to Mendeley reference manager. A screenshot of all references added to Mendeley is given in section 12, appendix A, as evidence for the use of reference management tools.

11. References

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12. Appendix A

AUTHORS	YEAR	TITLE	SOURCE	ADDED ▾	FILE
Capuano, German; Rimoli,…	2019	Smart finite elements: A novel machine learning application	Computer Methods …	6:38 PM	
		ml_fea		6:36 PM	
Fukami, Kai; Fukagata, Ko…	2023	Super-resolution analysis via machine learning: a survey for fluid flows	Theoretical and Co…	6:35 PM	
Fukami, Kai; Fukagata, Ko…	2023	Super-resolution analysis via machine learning: a survey for fluid flows	Theoretical and Co…	6:24 PM	
Obiols-Sales, Octavi; Vish…	2020	CFDNet: A deep learning-based accelerator for fluid simulations	Proceedings of the I…	6:08 PM	
Kumar, Tadbhagya; Pal, Pi…	2023	Development of a Machine Learning Wall Model for Large Eddy Simulation …	AIAA SciTech Foru…	5:21 PM	
Yang, X. I.A.; Zafar, S.; W…	2019	Predictive large-eddy-simulation wall modeling via physics-informed neural …	Physical Review Flu…	5:18 PM	
Yang, X. I.A.; Zafar, S.; W…	2019	Predictive large-eddy-simulation wall modeling via physics-informed neural …	Physical Review Flu…	5:18 PM	
Caron, Clément; Lauret, P…	2025	Machine Learning to speed up Computational Fluid Dynamics engineering …	Building and Enviro…	4:54 PM	
Capuano, German; Rimoli,…	2019	Smart finite elements: A novel machine learning application	Computer Methods …	4:54 PM	
Ageze, Mesfin Belayneh; …	2017	Wind Turbine Aeroelastic Modeling: Basics and Cutting Edge Trends	International Journa…	11/24/2024	
Khandan, Rasoul; Noroozi…	2012	The development of laminated composite plate theories: A review	Journal of Materials…	11/24/2024	
Bottasso, C. L.; Campagn…	2012	Multi-disciplinary constrained optimization of wind turbines	Multibody System D…	11/24/2024	
Blasques, José Pedro	2012	User's Manual for BECAS A cross section analysis tool for anisotropic and i…		11/24/2024	
Bottasso, C. L.; Campagn…	2014	Structural optimization of wind turbine rotor blades by multilevel sectional/m…	Multibody System D…	11/24/2024	
Bir, G; Migliore, P	2004	Preliminary Structural Design of Composite Blades for Two- and Three-Blad…		11/24/2024	
Hansen, M. O.L.; Sørense…	2006	State of the art in wind turbine aerodynamics and aeroelasticity	Progress in Aerosp…	11/24/2024	
Hansen, M. O.L.; Sørense…	2006	State of the art in wind turbine aerodynamics and aeroelasticity	Progress in Aerosp…	11/24/2024	
Lee, Hye Won; Roh, Myun…	2018	Review of the multibody dynamics in the applications of ships and offshore …	Ocean Engineering	11/24/2024	
Carrera Simone Ferrari Gi…	2019	POLYTECHNIC OF TURIN Structural Modelling and Aeroelastic Analysis of…		11/24/2024	
Jonkman, J M; Buhl, M L	2005	FAST User's Guide - Updated August 2005		11/24/2024	
Ageze, Mesfin Belayneh; …	2017	Wind Turbine Aeroelastic Modeling: Basics and Cutting Edge Trends	International Journa…	11/24/2024	
Fudlailah, Pratiwi; Allen, D…	2024	Verification of Euler–Bernoulli beam theory model for wind blade structure a…	Thin-Walled Structu…	11/24/2024	
Beck, André Teófilo; da Sil…	2011	Timoshenko versus Euler beam theory: Pitfalls of a deterministic approach	Structural Safety	11/24/2024	
Bauchau, Olivier A.; Han, …	2013	Advanced beam theory for multibody dynamics	Proceedings of the …	11/24/2024	
Bauchau, O. A.	1985	A beam theory for anisotropic materials	Journal of Applied …	11/24/2024	
Carrera Simone Ferrari Gi…	2019	POLYTECHNIC OF TURIN Structural Modelling and Aeroelastic Analysis of…		11/24/2024	
Bottasso, C. L.; Campagn…	2014	Structural optimization of wind turbine rotor blades by multilevel sectional/m…	Multibody System D…	11/24/2024	

Brown, Kenneth; Bortolotti…	2024	One-to-one aeroservoelastic validation of operational loads and performanc…	Wind Energy Science	11/23/2024
Batay, Sagidolla; Kamalov,…	2023	Adjoint-Based High-Fidelity Concurrent Aerodynamic Design Optimization o…	Fluids	11/23/2024
Faraggiana, Emilio; Giorgi,…	2022	A review of numerical modelling and optimisation of the floating support stru…	Journal of Ocean E…	11/23/2024
Sebastian Perez Becker, …		Advanced Aerodynamic Modeling and Control Strategies for Load Reductio…		11/23/2024
Faraggiana, Emilio; Giorgi,…	2022	A review of numerical modelling and optimisation of the floating support stru…	Journal of Ocean E…	11/23/2024
Zhong, W.; Shen, W. Z.; W…	2020	A tip loss correction model for wind turbine aerodynamic performance predi…	Renewable Energy	11/23/2024
Prandtl, By L		APPLICATIONS OF MODERN HYDRODYNAMICS TO AERONAUTICS. F…		11/23/2024
Mancini, Simone; Boorsm…	2023	A comparison of dynamic inflow models for the blade element momentum m…	Wind Energy Science	11/23/2024
Bortolotti, Pietro; Chetan, …	2024	Wind Turbine Aeroelastic Stability in OpenFAST	Journal of Physics: …	11/23/2024
Glauert, H.	1930	Aerodynamic Theory	The Journal of the …	11/23/2024
Sebastian Perez Becker, …		Advanced Aerodynamic Modeling and Control Strategies for Load Reductio…		11/22/2024
Men, Jiyuan; Ma, Gang; M…	2024	Aeroelastic instability analysis of floating offshore and onshore wind turbine…	Ocean Engineering	11/22/2024
Brown, Kenneth; Bortolotti…	2024	One-to-one aeroservoelastic validation of operational loads and performanc…	Wind Energy Science	11/22/2024
Bortolotti, Pietro; Chetan, …	2024	Wind Turbine Aeroelastic Stability in OpenFAST	Journal of Physics: …	11/22/2024
Zhang, Wenzhe; Calderon…	2024	Computational Fluid Dynamics (CFD) applications in Floating Offshore Win…	Applied Ocean Res…	11/21/2024
Sirigu, Massimo; Gigliotti, …	2024	The importance of aeroelasticity in estimating multiaxial fatigue behaviour o…	Heliyon	11/21/2024
Firoozi, Ali Akbar; Hejazi, …	2024	Advancing Wind Energy Efficiency: A Systematic Review of Aerodynamic O…	Energies	11/21/2024
Huang, Yang; Yang, Xiaol…	2024	Aeroelastic analysis of wind turbine under diverse inflow conditions	Ocean Engineering	11/20/2024
Wang, Lin; Liu, Xiongwei; …	2016	State of the art in the aeroelasticity of wind turbine blades: Aeroelastic mod…	Renewable and Sus…	11/20/2024
Deng, Xiao Wei; Wu, Nan;…	2019	Integrated design framework of next-generation 85-m wind turbine blade: M…	Composites Part B: …	11/20/2024
Wang, Lin; Liu, Xiongwei; …	2014	Nonlinear aeroelastic modelling for wind turbine blades based on blade ele…	Energy	11/20/2024

AUTHORS	YEAR	TITLE	SOURCE	ADDED	FILE
Bottasso, C. L.; Campagn…	2014	Structural optimization of wind turbine rotor blades by multilevel sectional/m…	Multibody System D…	11/24/2024	✓
Ageze, Mesfin Belayneh; …	2017	Wind Turbine Aeroelastic Modeling: Basics and Cutting Edge Trends	International Journa…	11/24/2024	✓
Rafiee, Roham; Tahani, M…	2016	Simulation of aeroelastic behavior in a composite wind turbine blade	Journal of Wind En…	11/24/2024	✓
Concli, F; Gorla, C		Analysis of the power losses in geared transmissions-measurements and C…		11/23/2024	✓
Ford, William	2015	Basic Iterative Methods	Numerical Linear Al…	11/23/2024	✓
NASA	2021	Navier Stokes Equations	NASA	11/23/2024	
Merle, A.; Bekemeyer, P.; …	2023	Adjoint high-dimensional aircraft shape optimization using a CAD-ROM par…	CEAS Aeronautical …	11/23/2024	✓
Xu, Shun; Xue, Yingjie; Zh…	2022	A Review of High-Fidelity Computational Fluid Dynamics for Floating Offsho…	Journal of Marine S…	11/23/2024	✓
Zhang, Shining; Ishihara, …	2018	Numerical study of hydrodynamic coefficients of multiple heave plates by lar…	Ocean Engineering	11/23/2024	✓
		RechAerosp_1994_SpalartAllmaras		11/23/2024	✓
Menter, Florian R		Improved Two-Equation k-Turbulence Models for Aerodynamic Flows		11/23/2024	✓
Menter, Florian R		Improved Two-Equation k-Turbulence Models for Aerodynamic Flows		11/23/2024	✓
Ferziger, Joel H; Perić, Mil…		Computational Methods for Fluid Dynamics		11/23/2024	✓
Houzeaux, G; Eguzkitza, …	2013	A Chimera Method for the Incompressible Navier-Stokes Equations	INTERNATIONAL J…	11/23/2024	✓
Rao, Mahesh J.; Nallayara…	2021	CFD approach to heave damping of spar with heave plates with experiment…	Applied Ocean Res…	11/23/2024	✓
Shourangiz-Haghighi, Alire…	2020	State of the Art in the Optimisation of Wind Turbine Performance Using CFD	Archives of Comput…	11/23/2024	✓
Men, Jiyuan; Ma, Gang; M…	2024	Aeroelastic instability analysis of floating offshore and onshore wind turbine…	Ocean Engineering	11/23/2024	✓
Bin Dissertation, BY LI		DYNAMIC MODELING, SIMULATION, AND CONTROL OF TRANSPORTA…		11/23/2024	✓
Huang, Yang; Yang, Xiaol…	2024	Aeroelastic analysis of wind turbine under diverse inflow conditions	Ocean Engineering	11/23/2024	✓
Wang, Lin; Liu, Xiongwei; …	2016	State of the art in the aeroelasticity of wind turbine blades: Aeroelastic mod…	Renewable and Sus…	11/23/2024	✓
Rinker, J M; Lund, R S	2023	General rights Verification of numerical lidars in HAWC2 Analysis of nacelle…		11/23/2024	✓
Zhang, Wenzhe; Calderon…	2024	Computational Fluid Dynamics (CFD) applications in Floating Offshore Win…	Applied Ocean Res…	11/23/2024	✓
Rinker, Jennifer; Gaertner,…	2020	Comparison of loads from HAWC2 and OpenFAST for the IEA Wind 15 MW…	Journal of Physics: …	11/23/2024	✓
Wang, Lin; Liu, Xiongwei; …	2014	Nonlinear aeroelastic modelling for wind turbine blades based on blade ele…	Energy	11/23/2024	✓
Deng, Xiao Wei; Wu, Nan;…	2019	Integrated design framework of next-generation 85-m wind turbine blade: M…	Composites Part B: …	11/23/2024	✓
Sirigu, Massimo; Gigliotti, …	2024	The importance of aeroelasticity in estimating multiaxial fatigue behaviour o…	Heliyon	11/23/2024	✓
Firoozi, Ali Akbar; Hejazi, …	2024	Advancing Wind Energy Efficiency: A Systematic Review of Aerodynamic O…	Energies	11/23/2024	✓
Brown, Kenneth; Bortolotti…	2024	One-to-one aeroservoelastic validation of operational loads and performanc…	Wind Energy Science	11/23/2024	✓